INEQUALITY ON NEF BASE LOCUS

For an effective Cartier divisor F on a smooth variety and an integral subvariety V, write $\mathrm{mult}_V(F)$ for the multiplicity of F at a general point of V. Equivalently, if f is a local equation of F at the generic point η_V, then

$$\mathrm{mult}_V(F) = \max\{j \geq 0 : f \in \mathfrak{m}_{\eta_V}^j\},$$

where $\mathfrak{m}_{\eta_V}$ is the maximal ideal of $\mathcal{O}_{X,\eta_V}$.

Lemma 1. *Let X be a smooth projective variety of dimension n, and let D be a nef Cartier divisor. Let $\Lambda \subseteq |D|$ be a nonempty linear system and V an integral component of its base locus, of codimension c. If, for an integer $h > 0$,*

$$\mathrm{mult}_V(F) \geq h \qquad \text{for every } F \in \Lambda,$$

then

$$h^c D^{n-c} \cdot V \leq D^n.$$

Proof. Intersect successively with c general members of Λ, discarding the components contained in the base locus before each further cut. The remaining components meet the next member properly. Since D is nef, discarding effective components does not increase the degree measured by the remaining powers of D. The resulting codimension-c cycle therefore has D-degree at most D^n.

At η_V, the base ideal is primary for the maximal ideal: this uses that V is a component of the base locus. No discarded component contains η_V. The c general local equations $f_1, \ldots, f_c$ form a parameter ideal I in the regular local ring $R = \mathcal{O}_{X,\eta_V}$ of dimension c. Its local intersection multiplicity is $\mathrm{length}(R/I)$. The hypothesis gives $I \subseteq \mathfrak{m}^h$, so monotonicity of Hilbert–Samuel multiplicity yields

$$\mathrm{length}(R/I) = e(I, R) \geq e(\mathfrak{m}^h, R) = h^c.$$

Thus the coefficient of V is at least h^c. Taking D-degree and using nefness for all remaining effective components proves the inequality. □

Corollary 2. *Let X, D, Λ be as in Lemma 1, and let x be an isolated point of the base locus of Λ. If $\mathrm{mult}_x(F) \geq r$ for every $F \in \Lambda$, where $r \geq 0$, then*

$$r^n \leq D^n.$$

Proof. For $r > 0$, apply Lemma 1 with $V = \{x\}$, $c = n$, and $h = \lceil r \rceil$. For $r = 0$, the assertion follows from nefness of D. □